

Trainings ▾

General Information

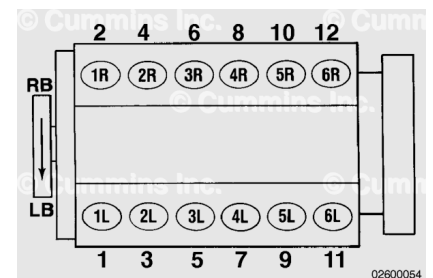
Valves **must** be adjusted correctly for the engine to operate efficiently. Valve adjustment **must** be performed using the values listed in this procedure.

Engine firing order: 1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L

Cylinders are numbered from the front gear cover end of the engine.

To determine the right and left banks on the engine, stand at the rear of the engine and face the front.

Two crankshaft revolutions are required to adjust all of the valves.



Each cylinder has two rocker levers. On the left bank, the lever nearest to the rear of the engine is the intake lever. On the right bank, the exhaust valve is nearest to the rear.

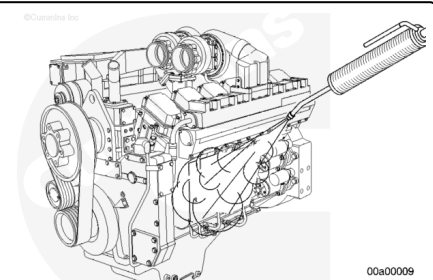
One pair of valves are adjusted at each index mark before rotating the engine to the next index mark.



⚠ WARNING ⚠

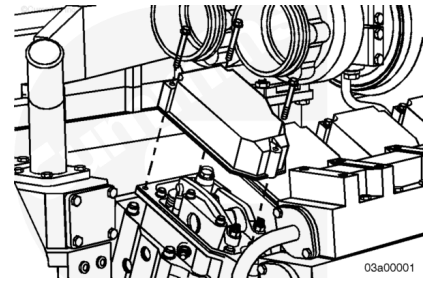
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

If you have **not** previously cleaned the engine, steam clean the engine to prevent dirt from entering the engine when the valve covers are removed. Refer to Section 5.

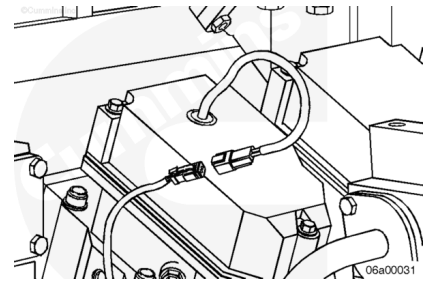


Remove

Remove the rocker lever covers and all related components.

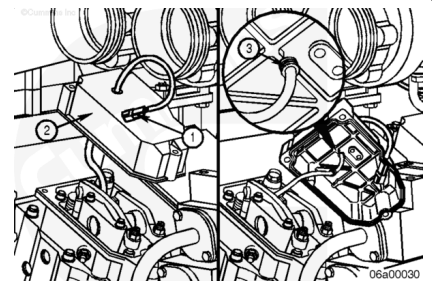


Disconnect the NBF injector lead from the wiring harness.



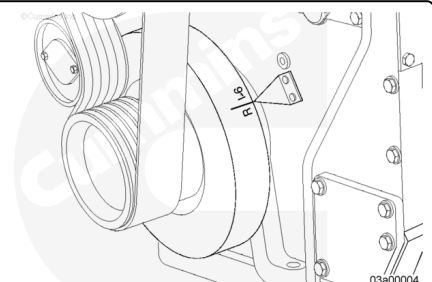
Remove the rubber grommet from the rocker lever cover.

Pass the NBF lead connector through the hole in the rocker lever cover.

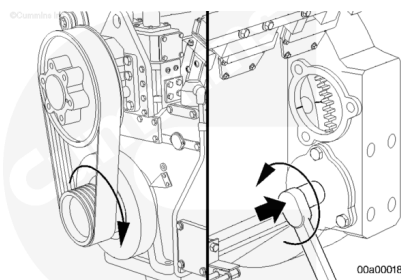


Adjust

Valve adjustment marks are on the vibration damper. The marks **must** be aligned with the pointer.

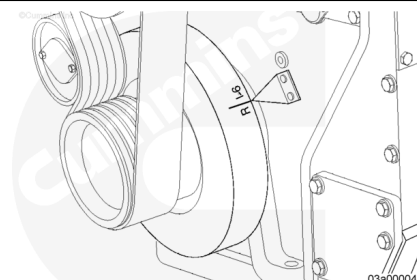


Rotate the engine **clockwise**, as the arrow on the pulley indicates, with a barring tool.



Check that the number 1 right-bank valves are ready to be adjusted by seeing that both rocker levers on the number 1 right bank are even and both rocker levers on the number 6 right bank are uneven.

If **not**, rotate the engine 180 degrees.



QST30 Valve Adjustment Chart

Vibration Damper Mark	Valves Closed or Push Rod Protrusion Even on Cylinder Number	Adjust Valves on Cylinder Number
R1-6	1RB	1RB
L1-6	1LB	1LB
R2-5	5RB	5RB
L2-5	5LB	5LB
R3-4	3RB	3RB
L3-4	3LB	3LB
R1-6	6RB	6RB
L1-6	6LB	6LB
R2-5	2RB	2RB
L2-5	2LB	2LB
R3-4	4RB	4RB
L3-4	4LB	4LB

The rocker levers will be the same height on the cylinder ready for valve adjustment.

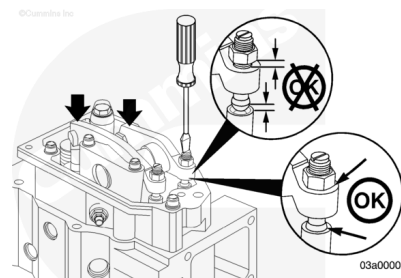
The crossheads and valves are ready to be adjusted on the cylinder that has all the valves closed.

Check the two cylinders shown by a mark on the vibration damper.

If the rocker lever assemblies have been removed, use this step to determine the cylinder to set.

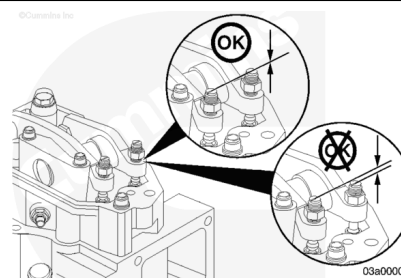
All adjusting screws **must** be loose on all cylinders and the push rods **must** remain in alignment.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.

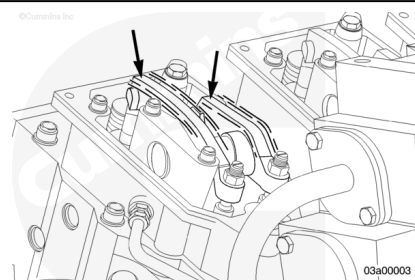


The cylinder with the adjusting screws that are nearly the same height is ready for valve adjustment. The second cylinder that is **not** ready for adjustment will have the adjusting screw for the intake valves more than five threads above the exhaust screw.

The rocker levers will be close to the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.



If the rocker levers have **not** been removed and installed again, wiggle the valve rocker levers on the two cylinders in question. The crossheads and valves on the cylinder where both levers feel loose are ready to adjust.



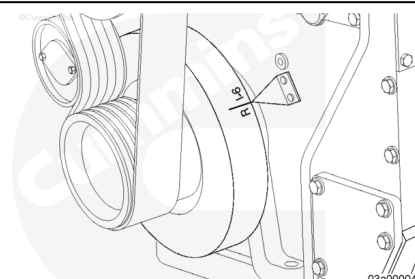
⚠ CAUTION ⚠

Use the correct firing order for the engine. Failure to do so can cause engine damage.

Adjustment can begin on any valve set mark. In the example, assume the R1-6 mark is aligned and the adjusting screw height for the valve on cylinder number 1 right bank is the same, indicating the valve is closed and ready to adjust.

After identifying which cylinder is ready to be adjusted, follow the engine firing order for subsequent adjustments.

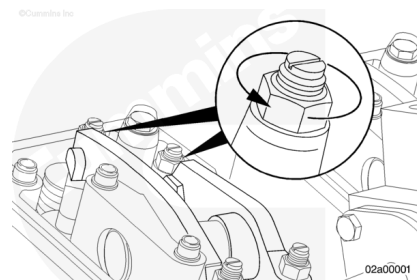
The firing order for the engine is 1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L.



Crosshead adjustment **must always** be made before attempting to adjust the valves.

Adjust the crossheads on the cylinder that has both valves closed.

Loosen the crosshead adjusting screw locknuts on the intake and exhaust valve crossheads.

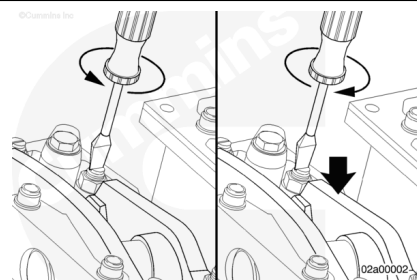


Use the following procedure to adjust both the intake and exhaust crossheads.

Turn the adjusting screw out at least one turn.

Hold the crosshead down against its guide.

Turn the adjusting screw in until it touches the top of the valve stem but does **not** raise the crosshead.



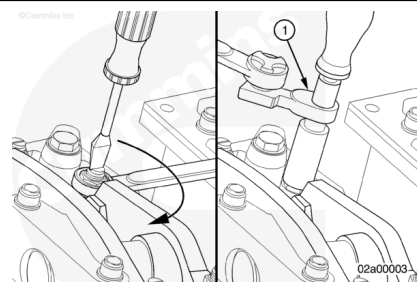
The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the locknut using torque wrench service tool, Part Number ST-699, with adapter (1).

Torque Value: 45 n•m [35 ft-lb]

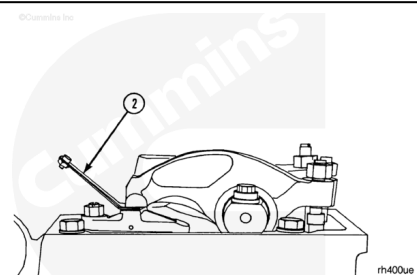
Tighten the locknut using torque wrench service tool, Part Number ST-699, without adapter.

Torque Value: 60 n•m [45 ft-lb]



Select a feeler gauge for the correct valve lash specification.

Insert the gauge (2) between and rocker lever and the crosshead.

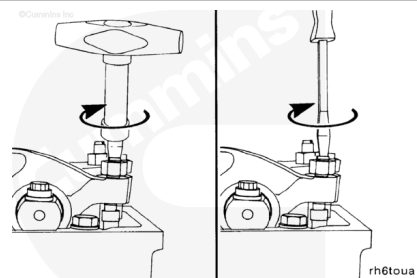


Intake and Exhaust Valve Adjustment

Valve	mm	[in]
Exhaust	0.80	0.032
Intake	0.43	0.017

Two different methods for establishing valve lash clearance are described below. The two methods are torque wrench method or feel method; either method can be used; however, the torque wrench method has proven to be the most consistent.

Use a service tool inch pound torque wrench, Part Number 3376592.

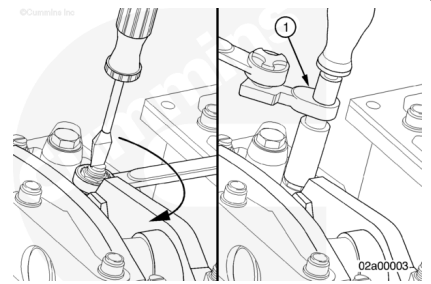


Torque Value: 0.68 n•m [6 in-lb]

For the feel method, use a screwdriver to turn the adjusting screw **only** until the lever touches the feeler gauge.

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the locknut using torque wrench service tool, Part Number ST-699, with adapter (1).



Torque Value: 45 n•m [35 ft-lb]

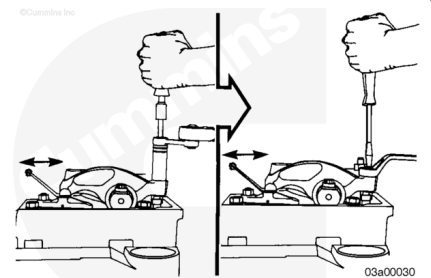
Tighten the locknut using torque wrench service tool, Part Number ST-699, without adapter.

Torque Value: 60 n•m [45 ft-lb]

The feeler gauge **must** slide backward and forward with **only** a slight drag.

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

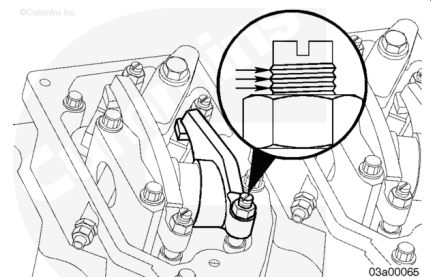
Repeat the adjustment process until the clearance is correct on both the intake and the exhaust valves on the cylinder being adjusted.



After the overhead has been completely set, check the adjusting screw protrusion. Approximately three threads **must** be exposed above the locknut.

If **not**, look for:

- Overhead set improperly
- Rocker housing gasket missing
- Push rod **not** in socket
- Cam follower socket **not** seated properly
- Nut thickness **not** standard
- Crosshead off valve stem.



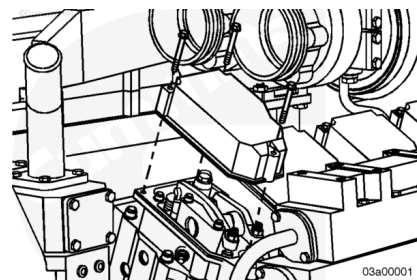
Install

Install each rocker cover gasket, rocker lever cover, and capscrews.

Tighten all the capscrews.

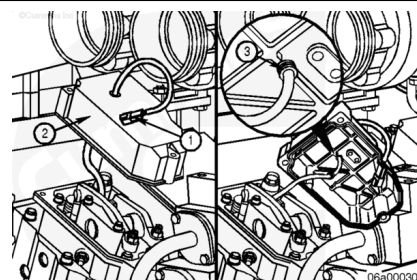
Torque Value: 5 n•m [44 in-lb]

Note : For industrial engines, the number 1 rocker cover on both the right and left banks has a hole for a grommet, for the NBF injector lead.



Install the rubber grommet from the rocker lever cover.

Pass the NBF lead connector through the hole in the rocker lever cover.



Connect the NBF injector lead from the wiring harness.

